

SciLab (Lab 02)

Abdon Morales

June 5, 2026

Description

This lab will familiarize the student with the capabilities and basic use of SciLab which is a freely available calculation program. It is similar to the much more expensive commercial tool MATLAB. MATLAB is widely used in industry and has advanced capabilities in graphics plotting and code generation that go well beyond the capabilities of SciLab.

This psuedo code just describes the math transforms necessary to implement

1. Polar to Cartesian transforms and
2. Cartesian to polar transforms

Calculations

```
%j=%i
function x=polar2cart(m, a);
x=m*cos(a)+(%j*m*sin(a));
endfunction

function p=cart2polar(x);
re = real(x); im = imag(x);
m = sqrt(re^2 + im^2);
a = atan(im, re);
p = [m,a];
endfunction

// ----- MAIN PROGRAM -----
disp("Abdon Morales");
disp("ENGR 2405 - Polar/Cartesian Conversions Testcases");
disp("=====");
// --- Test cart2polar ---
disp("--- Testing cart2polar ---");
x1 = 3 + 4*i;
printf("Input (cartesian): %g + %gi\n", real(x1), imag(x1));
out1 = cart2polar(x1);
printf("Output [m a]: m = %g, a = %g rad (%g deg)\n", ...
    out1(1), out1(2), out1(2)*180/pi);
// --- Test polar2cart ---
disp("--- Testing polar2cart ---");
m2 = 5;
a2 = 0.9273; // ~53.13 deg expressed in radians
printf("Input (polar): m = %g, a = %g rad\n", m2, a2);
out2 = polar2cart(m2, a2);
printf("Output (cartesian): %g + %gi\n", real(out2), imag(out2));
// --- Round-trip check ---
```

```

disp("--- Round-trip: cart -> polar -> cart ---");
back = polar2cart(out1(1), out1(2));
printf("Should match 3 + 4i:  %g + %gi\n", real(back), imag(back));
// --- Quadrant test: -3 + 4i (second quadrant) ---
disp("--- Testing cart2polar in Quadrant II ---");
x3 = -3 + 4*i;
printf("Input  (cartesian): %g + %gi\n", real(x3), imag(x3));
out3 = cart2polar(x3);
printf("Output [m a]:  m = %g,  a = %g rad  (%g deg)\n", ...
       out3(1), out3(2), out3(2)*180/%pi);

```

Listing 1: SciLab Part I

```

"Abdon Morales"
"ENGR 2405 - Polar/Cartesian Conversions Testcases"
"=====
--- Testing cart2polar ---"
Input  (cartesian): 3 + 4i
Output [m a]:  m = 5,  a = 0.927295 rad  (53.1301 deg)
--- Testing polar2cart ---"
Input  (polar): m = 5,  a = 0.9273 rad
Output (cartesian): 2.99998 + 4.00001i
--- Round-trip: cart -> polar -> cart ---"
Should match 3 + 4i:  3 + 4i
--- Testing cart2polar in Quadrant II ---"
Input  (cartesian): -3 + 4i
Output [m a]:  m = 5,  a = 2.2143 rad  (126.87 deg)

```

Listing 2: SciLab Part I runtime output

```

// ----- MAIN PROGRAM (Part II) -----
disp("Abdon Morales");
disp("ENGR 2405 - System of Linear Equations");
disp("=====");
// Build the matrix and vector
A = [2 1 1; 1 2 1; 1 1 2];
X = [1;2;3];
disp("Matrix A = "); disp(A);
disp("Vector X = "); disp(X);

// Do the math via matrix-vector multiplication
Y = A*X;
disp("This is the result of A*X");
disp(Y);

// Do the matrix A inversion
invA = inv(A);
disp("This is A inverse =");
disp(invA);

// Now go back to find X using the inversion
out = invA * Y;
disp("Vector X using A inverse * Y");
disp(out);

```

Listing 3: SciLab Part II

```
"Abdon Morales"
"ENGR 2405 - System of Linear Equations"
"=====
"Matrix A = "
  2.   1.   1.
  1.   2.   1.
  1.   1.   2.
"Vector X = "
  1.
  2.
  3.
"This is the result of A*X"
  7.
  8.
  9.
"This is A inverse ="
  0.75  -0.25  -0.25
 -0.25   0.75  -0.25
 -0.25  -0.25   0.75
"Vector X using A inverse * Y"
  1.
  2.
  3.
```

Listing 4: SciLab Part II runtime output

Observations & Discussion

None.